

Lecture 10: Model Selection and Validation

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September 28, 2026



Introduction

- In earlier chapters, we considered how to fit simple and multiple regression models and how to make inferences from these models.
- In this chapter, we present an overview of the model-building, model selection and model-validation.

Outline

- 1 Overview of model-building process
- 2 Surgical unit example
- 3 Criteria for Model Selection
- 4 Automatic search procedures for model selection
- 5 Model validation

Outline

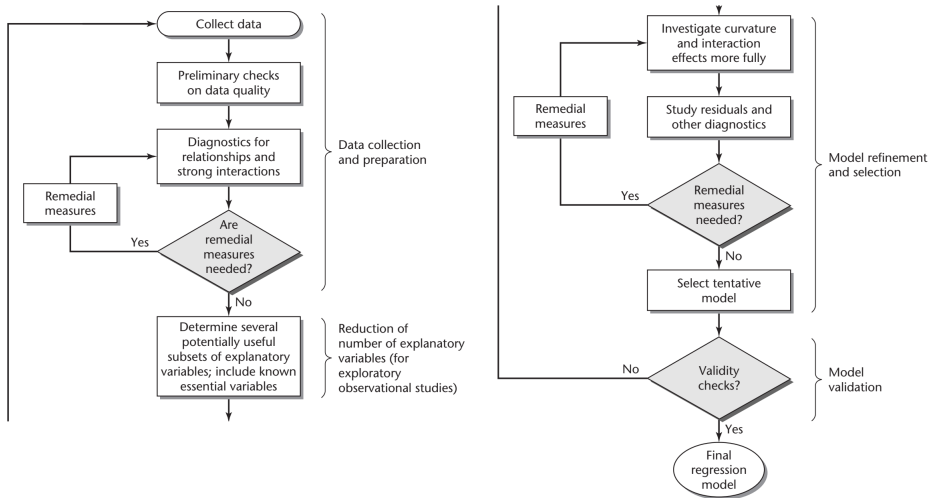
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Strategy for the building of a regression model

A strategy for the building of a regression model includes three or, sometimes, four phases:

- ① Data collection and preparation
- ② Reduction of explanatory or predictor variables (for exploratory observational studies)
- ③ Model refinement and selection
- ④ Model validation

Strategy for the building of a regression model



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Surgical unit example

A hospital surgical unit was interested in predicting survival in patients undergoing a particular type of liver operation. A random selection of 108 patients was available for analysis. From each patient record, the following information was extracted from the preoperation evaluation:

- X_1 : blood clotting score
- X_2 : prognostic index
- X_3 : enzyme function test score
- X_4 : liver function test score
- X_5 : age, in years
- X_6 : indicator variable for gender (0 for male, 1 for female)
- X_7 and X_8 : indicator variables for history of alcohol use:
 $(X_7, X_8) = (0, 0), (1, 0)$ and $(0, 1)$ for none, moderate and severe use, respectively.
- Y : survival time

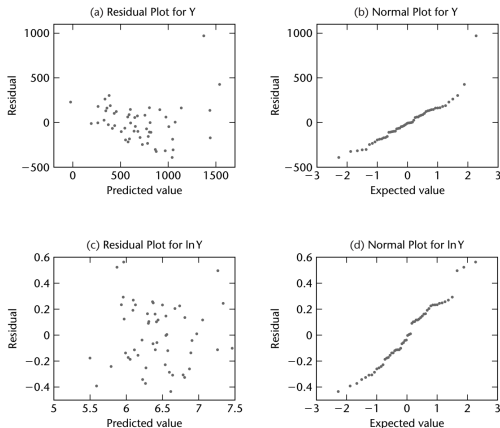
Surgical unit example

- Since the pool of predictor variables is small, a reasonably full exploration of relationships and of possible strong interaction effects is possible at this stage of data preparation
 - Stem-and-leaf plots can be used to highlight several cases as outlying
 - A scatter plot matrix and the correlation matrix were also obtained

Surgical unit example

First model: first-order regression model based on all predictor variables was fitted.

Second model: logarithmic transformation $Y' = \ln Y$

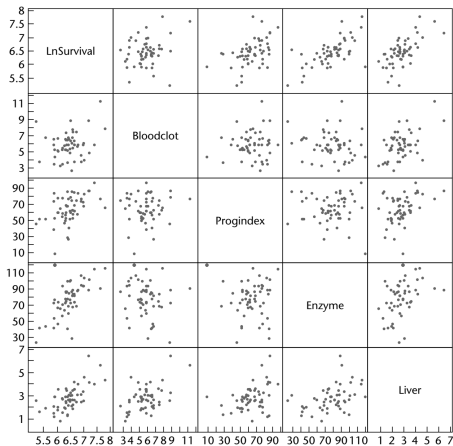


Surgical unit example

Multivariate Correlations

	LnSurvival	Bloodclot	Progindex	Enzyme	Liver
LnSurvival	1.0000	0.2462	0.4699	0.6539	0.6493
Bloodclot	0.2462	1.0000	0.0901	-0.1496	0.5024
Progindex	0.4699	0.0901	1.0000	-0.0236	0.3690
Enzyme	0.6539	-0.1496	-0.0236	1.0000	0.4164
Liver	0.6493	0.5024	0.3690	0.4164	1.0000

Scatterplot Matrix



- X4 has moderately high pairwise correlations with X1, X2, and X3

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Model selection

- If there are 10 variables, we have 2^{10} possible models. TOO MANY!
- We focus on 06 values R_p^2 , $R_{a,p}^2$, C_p , AIC_p , SBC_p , and $PRESS_p$ for model selection procedures.
- **Denote:** $P - 1$ be total number of variables and $p - 1$ be the number of variables in the model (p parameters).

TABLE 9.2 SSE_p , R_p^2 , $R_{a,p}^2$, C_p , AIC_p , SBC_p , and $PRESS_p$ Values for All Possible Regression Models—Surgical Unit Example.

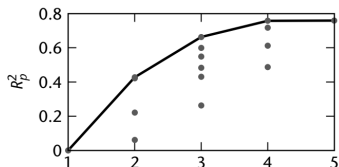
X Variables in Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	p	SSE_p	R_p^2	$R_{a,p}^2$	C_p	AIC_p	SBC_p	$PRESS_p$
None	1	12.808	0.000	0.000	151.498	-75.703	-73.714	13.296
X_1	2	12.031	0.061	0.043	141.164	-77.079	-73.101	13.512
X_2	2	9.979	0.221	0.206	108.556	-87.178	-83.200	10.744
X_3	2	7.332	0.428	0.417	66.489	-103.827	-99.849	8.327
X_4	2	7.409	0.422	0.410	67.715	-103.262	-99.284	8.025
X_1, X_2	3	9.443	0.263	0.234	102.031	-88.162	-82.195	11.062
X_1, X_3	3	5.781	0.549	0.531	43.852	-114.658	-108.691	6.988
X_1, X_4	3	7.299	0.430	0.408	67.972	-102.067	-96.100	8.472
X_2, X_3	3	4.312	0.663	0.650	20.520	-130.483	-124.516	5.065
X_2, X_4	3	6.622	0.483	0.463	57.215	-107.324	-101.357	7.476
X_3, X_4	3	5.130	0.599	0.584	33.504	-121.113	-115.146	6.121
X_1, X_2, X_3	4	3.109	0.757	0.743	3.391	-146.161	-138.205	3.914
X_1, X_2, X_4	4	6.570	0.487	0.456	58.392	-105.748	-97.792	7.903
X_1, X_3, X_4	4	4.968	0.612	0.589	32.932	-120.844	-112.888	6.207
X_2, X_3, X_4	4	3.614	0.718	0.701	11.424	-138.023	-130.067	4.597
X_1, X_2, X_3, X_4	5	3.084	0.759	0.740	5.000	-144.590	-134.645	4.069

R_p^2 and SSE_p criterion

- R_p^2 : coefficient of multiple determination with $p - 1$ variables in the model

$$R_p^2 = 1 - \frac{SSE_p}{SSTO}$$

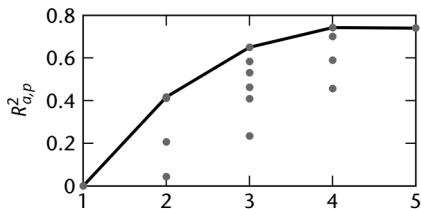
- Adding more variable to the model always increases R_p^2 .



After 3 variables, there is small increase in R_p^2 . From Table 9.2, subset X_1, X_2, X_3 is a reasonable choice.

- **Adjusted** coefficient of multiple determination $R_{a,p}^2$:

$$R_{a,p}^2 = 1 - \frac{MSE_p}{SSTO/(n-1)} = 1 - \frac{n-1}{n-p} \frac{SSE_p}{SSTO}$$



Same suggestion: X_1, X_2, X_3 is a reasonable subset. 04 variables has a smaller adjusted R-square coefficient.

Mallows' C_p criterion

- This criterion is concerned with the total mean squared error of the n fitted values for each subset regression model.

$$C_p = \frac{SSE_p}{MSE(X_1, \dots, X_{p-1})} - (n - 2p)$$

- We seek to identify subsets of X variables for which (1) the C_p value is small and (2) the C_p value is near p .

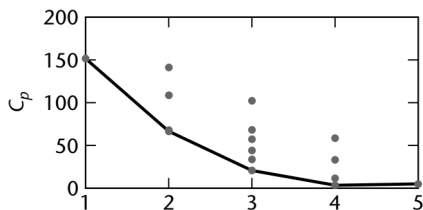


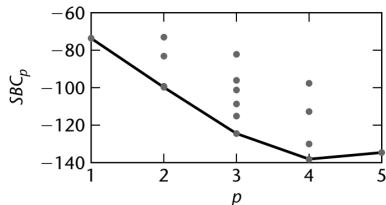
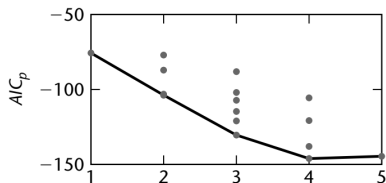
Figure and Table 9.2 show that X_1, X_2, X_3 is the best subset.

AIC_p and SBC_p criteria

- Both $R_{a,p}^2$ and C_p are model selection criteria that penalize models having large numbers of predictors.
- Two alternatives also provide penalties for adding predictors are Akaike's Information Criterion (AIC_p) and Schwarz' Bayesian SBC_p (a.k.a Bayesian Information Criterion - BIC).

$$AIC_p = n * \ln (SSE_p/n) + 2p$$

$$SBC_p = n * \ln (SSE_p/n) + p * \ln (n)$$



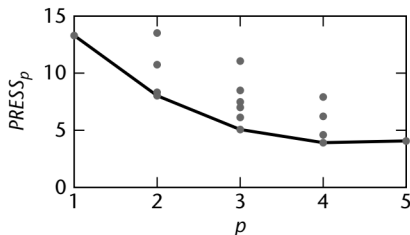
Both of these criteria are minimized for subset (X_1, X_2, X_3) .

$PRESS_p$ Criterion

- $PRESS_p$ (prediction sum of squares) criterion is a measure of how well the use of the fitted values for a subset model can predict the observed responses Y_i
- Let $\hat{Y}_{i(i)}$: fitted value of i th data point using the model in which i is omitted.

$$PRESS_p = \sum_{i=1}^n (Y_i - \hat{Y}_{i(i)})^2.$$

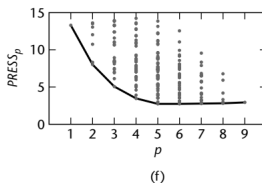
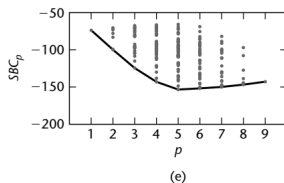
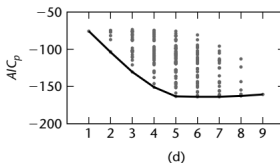
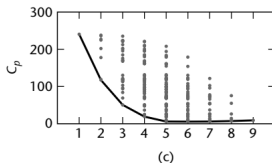
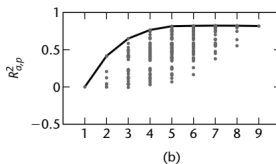
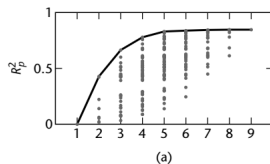
- Models with small $PRESS_p$ values are considered good candidate models.



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Best subset algorithms



Best subset algorithms: exhausted search for all 9 variables.

Best subset algorithms

p	(1) SSE_p	(2) R_p^2	(3) $R_{a,p}^2$	(4) C_p	(5) AIC_p	(6) SBC_p	(7) $PRESS_p$
1	12.808	0.000	0.000	240.452	-75.703	-73.714	13.296
2	7.332	0.428	0.417	117.409	-103.827	-99.849	8.025
3	4.312	0.663	0.650	50.472	-130.483	-124.516	5.065
4	2.843	0.778	0.765	18.914	-150.985	-143.029	3.469
5	2.179	0.830	0.816	5.751	-163.351	<u>-153.406</u>	<u>2.738</u>
6	2.082	0.837	0.821	<u>5.541</u>	-163.805	-151.871	2.739
7	2.005	0.843	<u>0.823</u>	5.787	<u>-163.834</u>	-149.911	2.772
8	1.972	0.846	<u>0.823</u>	7.029	-162.736	-146.824	2.809
9	<u>1.971</u>	<u>0.846</u>	0.819	9.000	-160.771	-142.870	2.931

Best variable selection criterion values - surgical unit example.

Stepwise regression methods

For large number of variables, we can use stepwise methods.

- forward selection: start from simple linear regression and select the variable with the largest t^* statistic.
- backward elimination: start from the full model and remove the variable with the largest p value.

Stepwise methods are available in R and some other not-free statistic softwares.

Lasso is another approach that will be introduced later.

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Model validation usually involves checking a candidate model against independent data. Three basic ways of validating a regression model are:

- Collection of new data to check the model and its predictive ability.
- Comparison of results with theoretical expectations, earlier empirical results, and simulation results.
- Use of a holdout sample to check the model and its predictive ability.

Collection of new data to check model

- One validation method is to reestimate the model form chosen earlier using the new data.
- A second validation method is designed to calibrate the predictive capability of the selected regression model.

Collection of new data to check model - Difficulties in Replicating a Study

- Difficulties often arise when new data are collected to validate a regression model, especially with observational studies.
- Even with controlled experiments, for instance, the laboratory equipment for the new study to be conducted in a different laboratory may differ from that used in the initial study, resulting in somewhat different calibrations for the response measurements
- the social sciences where controlled experiments often are not feasible. Repetition of an observational study usually involves different conditions, the differences being related to changes in setting and/or time.

Comparison with Theory, Empirical Evidence, or Simulation Results

- In some cases, theory, simulation results, or previous empirical results may be helpful in determining whether the selected model is reasonable.
- Comparisons of regression coefficients and predictions with theoretical expectations, previous empirical results, or simulation results should be made
- Unfortunately, there is often little theory that can be used to validate regression models.

Data splitting

- **Cross-validation:** when the data set is large enough is to split the data into two sets:
 - *model-building set* or the *training sample*, is used to develop the model.
 - *validation or prediction set*, is used to evaluate the reasonableness and predictive ability of the selected model.
- The regression coefficients can be reestimated for the selected model and then compared for consistency with the coefficients obtained from the model-building data set.
- Data sets are often split equally into model-building and validation data sets.
- It is important that the model-building data set be sufficiently large so that a reliable model can be developed (for instance, the number of cases should be at least 6 to 10 times the number of variables in the pool of predictor variables.)
- If the entire data set is not large enough under these circumstances for making an equal split, the validation data set will need to be smaller than the model-building data set.

Data splitting

- Splits of the data can be made at random
- Another possibility is to match cases in pairs and place one of each pair into one of the two split data sets
- When data are collected sequentially in time, it is often useful to pick a point in time to divide the data
- When seasonal or cyclical effects are present in the data (e.g., sales data), the split should be made at a point where the cycles are balanced

- Use of time or some other characteristic of the data to split the data set provides the opportunity to test the generalizability of the model since conditions may differ for the two data sets
- A possible drawback of data splitting is that the variances of the estimated regression coefficients developed from the model-building data set will usually be larger than those that would have been obtained from the fit to the entire data set.
- However, if the model-building data set is reasonably large, these variances generally will not be that much larger than those for the entire data set

- Chapter 9, Kutner's book
- STATS191, Introduction to Applied Statistic, Stanford